

Predicting Telecommunications Customer Churn

Design and Creative Technologies

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1. Problem and approach

Customer churn is costly because replacing a subscriber generally requires more investment than retaining one. Using the IBM/Kaggle Telco sample, I removed the five attributes specified in the brief, producing 7,043 customers and 16 attributes with *Churn* as the target (Kaggle, 2020). I built an interpretable decision tree in Spark MLlib and compared it with recall-focused alternatives (Apache Spark, 2024).

Figure 1 follows the analytics lifecycle: prepare the data, fit preprocessing only on training data, tune on validation, evaluate once on test, and translate performance into a retention decision (EMC Education Services, 2015). Deterministic stratification produced 4,225 training, 1,409 validation and 1,409 test customers with no overlapping IDs. This prevents held-out customers from influencing category indexes, feature selection or hyperparameters.

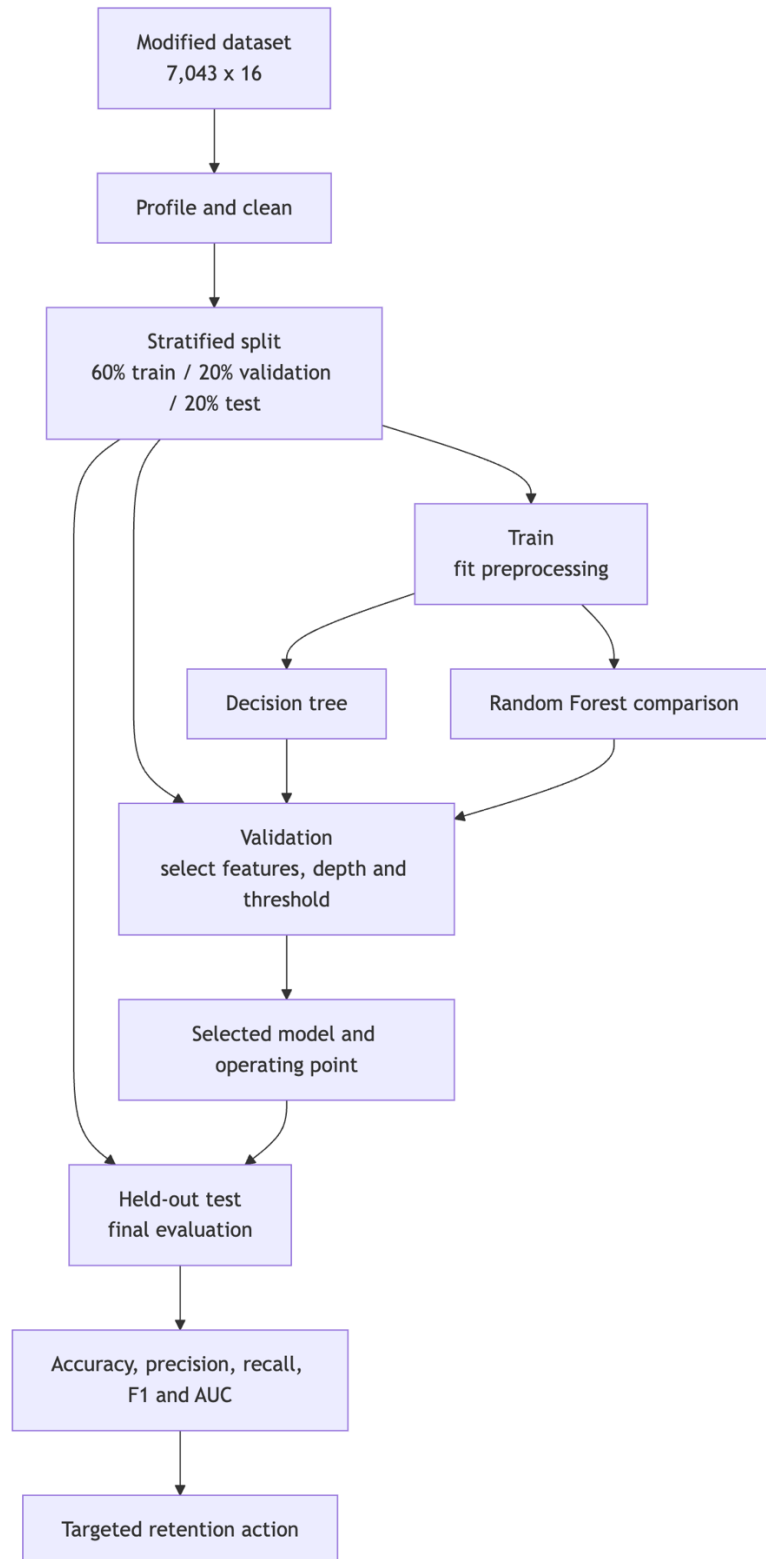


Figure 1. Leakage-safe churn-analysis workflow.

2. Exploration, cleaning and feature selection

The EDA supplied central-tendency and dispersion statistics plus bar charts, histograms, box plots, a heatmap and pair plot. Churn is imbalanced: 26.5% leave, so predicting that everyone stays already achieves 73.5% accuracy. Churn is concentrated among low-tenure, month-to-month customers without technical support. Median tenure was 10 months among churners and 38 among customers who stayed; *SeniorCitizen* was treated as binary categorical, not continuous.

Quality checks found no duplicate rows or customer IDs. The 11 blank *TotalCharges* records all had *tenure* = 0; they were set to zero because these new customers had accumulated no charges, a domain-informed correction preferable to a global median (Han et al., 2012). IQR checks found no outliers requiring removal. Even if an extreme had been found, it would require investigation rather than automatic deletion because a high-value or long-tenure customer may be legitimate.

Although *tenure* and *TotalCharges* were strongly correlated ($r = 0.826$), correlation alone did not decide removal. Validation ablation showed that dropping *TotalCharges* changed AUC from 0.8084 to 0.8068 and churn-F1 from 0.5223 to 0.5200. This negligible loss was accepted to remove redundancy and simplify the final 13-predictor model (Figure 2).

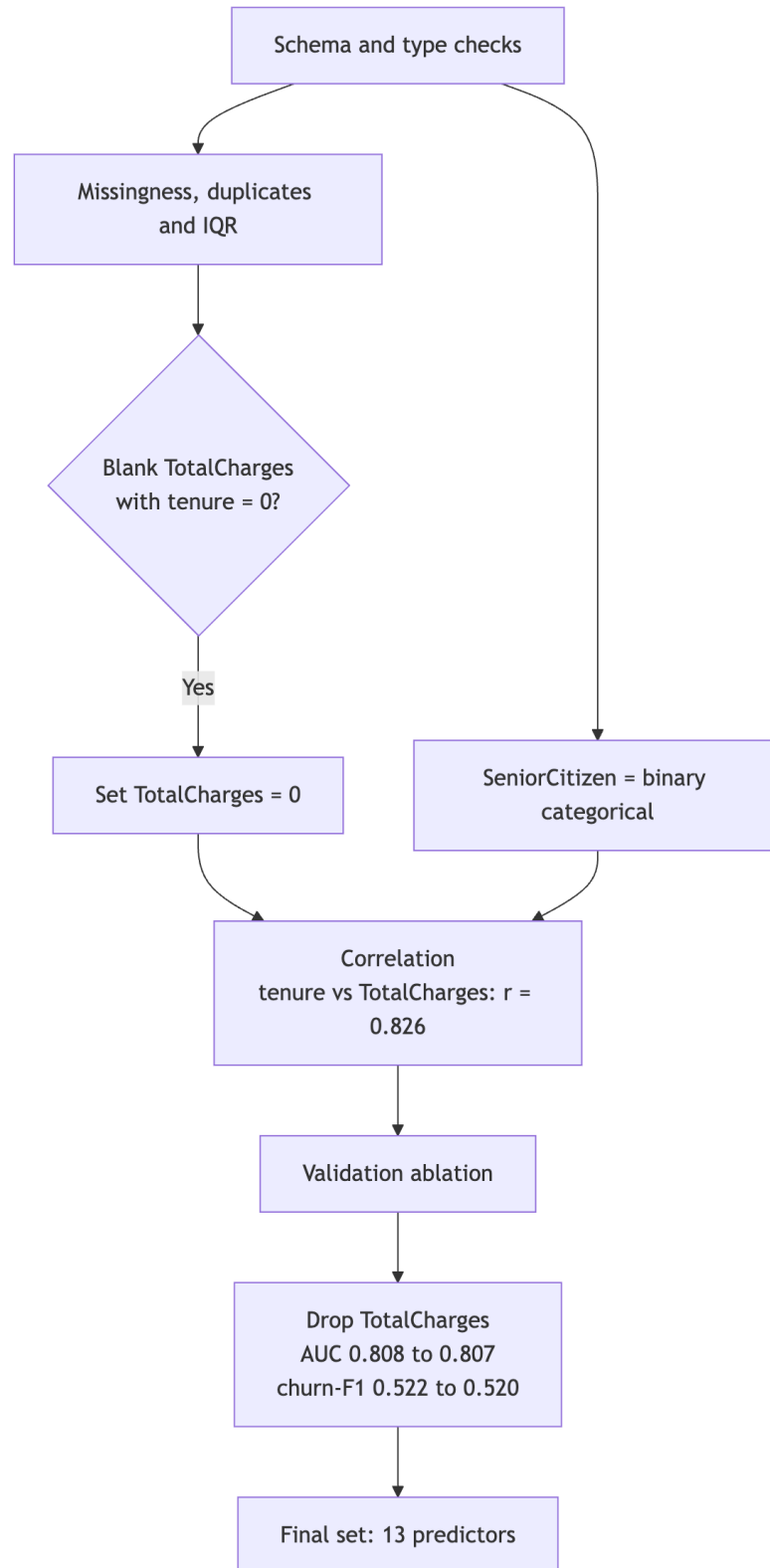


Figure 2. Data-quality and feature-selection decisions.

3. Missing-value strategy

The tree identified *Contract* as most important (importance 0.564), followed by *tenure* (0.168) and *TechSupport* (0.160). I simulated 30% missingness in *Contract*, derived its replacement mode (*Month-to-month*) from training only, and applied it unchanged to validation and the same held-out test customers. Accuracy moved from 0.781 to 0.779, while churn-F1 fell from 0.548 to 0.485. Stable accuracy therefore masked information loss in the minority class.

Mode imputation is reproducible and retains all rows, but the synthetic mask represents missing completely at random. Production gaps could instead depend on sales channel or customer group. A deployed pipeline should monitor missingness by source and compare an explicit missing category or model-based imputation. C4.5 fractional instances are a theoretical alternative (Witten et al., 2017), not Spark MLlib's implemented behavior.

4. Interpretation of churn analysis

4.1. Effectiveness and generalisation

The held-out tree classified 78.1% correctly, only 4.6 percentage points above the naive baseline. It caught 187 of 374 churners: recall 0.500, precision 0.605 and churn-F1 0.548. Its confusion matrix contained 913 true negatives, 122 false positives, 187 false negatives and 187 true positives. Accuracy is therefore unsuitable as the sole measure because half the retention opportunity remains undetected.

Validation selected *maxDepth* = 6, which acts as pre-pruning by stopping unnecessary recursive splits. Train-to-test gaps were modest: 1.79 percentage points in accuracy, 3.75 in churn-F1 and 2.23 in AUC (Appendix B). This suggests limited rather than severe overfitting. Test AUC was 0.802, well above the 0.500 expected from random ranking, although the default classification threshold still missed many churners. AUC and recall therefore answer different questions: ranking quality can be useful even when the selected operating point is poor.

4.2. Who is churning

The tree and EDA agree, and because a tree is a rule learner its branches read directly as retention rules that a team can apply without the model:

- **Rule 1 (high risk).** If Contract = Month-to-month AND TechSupport = No AND tenure ≤ 15.5 months, flag for retention. This segment holds 1,516 customers and churns at 60.0% against the 26.5% base rate; adding PaymentMethod = Electronic check tightens it to 914 customers at 66.2%.
- **Rule 2 (low risk).** If Contract is one- or two-year, spend no retention budget. This segment holds 3,168 customers and churns at 6.8%, and only 2.8% on two-year contracts.

The rules bound the retention list: the practical response is an early-tenure programme that converts month-to-month customers to longer commitments and bundles technical support, rather than indiscriminate discounts.

4.3.Improving detection

Validation threshold tuning moved the tree cut-off from 0.50 to 0.26, raising test recall from 0.500 to 0.685. Inverse-frequency weighting raised recall to 0.797. A cross-validated Random Forest ranked customers best (AUC 0.833); at threshold 0.30 it achieved recall 0.754 and churn-F1 0.618 (Table 1). Weighting changes the training objective, threshold tuning changes the decision rule, and the forest improves ranking by combining trees. The decision tree remains the required explanatory model; the forest is the stronger operational candidate. Threshold tuning leaves AUC unchanged because it does not reorder customer risk scores; it only changes where predictions switch from stay to churn.

Table A1 - Held-out comparison of the baseline decision tree and recall-focused alternatives.

Model	Threshold	Accuracy	Precision	Recall	Churn-F1	AUC
Decision tree baseline	0.50	0.781	0.605	0.500	0.548	0.802
Decision tree tuned	0.26	0.750	0.685	0.685	0.593	0.802
Weighted tree	0.50	0.725	0.489	0.797	0.606	0.809
Random Forest baseline	0.50	0.794	0.651	0.479	0.552	0.833
Random Forest tuned	0.30	0.753	0.524	0.754	0.618	0.833

4.4.Business operating point and responsible use

Because financial costs were unavailable, false-negative to false-positive ratios of 2:1, 5:1 and 10:1 were sensitivity scenarios, not measured facts (Appendix A). At 5:1, validation selected a Random Forest threshold of 0.14. Test recall reached 0.933, but precision fell to 0.418 and 486 false alarms were generated (Figure 3). The business budget owner must validate this trade-off against incentive cost and customer lifetime value.

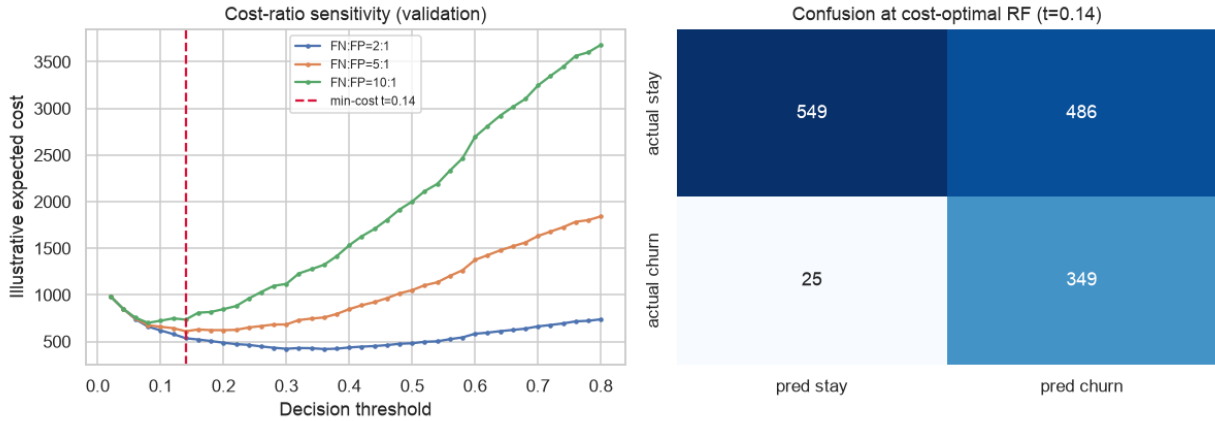


Figure 3. Cost-ratio sensitivity and held-out confusion matrix at the illustrative 5:1 operating point.

The demographic audit does not establish fairness, but it identifies monitoring needs. Female and male recall were similar (0.508 and 0.492), while false-positive rates were 0.141 and 0.096. Senior and non-senior rates were 0.162 and 0.112, respectively, with different churn prevalence and group sizes (Appendix C). These descriptive gaps cannot be attributed solely to the model because support and underlying churn rates differ. Deployment should monitor subgroup error rates over time and investigate data, threshold and service-process causes before using scores for customer treatment.

5. Conclusion

The decision tree provides an interpretable churn profile but misses half the churners at its default threshold. Leakage-safe preprocessing, evidence-based feature reduction and paired missing-value evaluation make that conclusion defensible. Threshold tuning, weighting and Random Forest improve detection, but the operating point must balance recovered customers

against wasted offers. A pilot should monitor churn recall, offer acceptance, missingness, subgroup errors and model drift before full deployment.

6. Appendices

6.1. Appendix A – Cost sensitivity

Table A1 - Random Forest sensitivity to illustrative false-negative and false-positive cost ratios.

FN:FP ratio	Threshold	TN	FP	FN	TP	Recall	Precision	Accuracy
2:1	0.36	838	197	124	250	0.668	0.559	0.772
5:1	0.14	549	486	25	349	0.933	0.418	0.637
10:1	0.08	379	656	9	365	0.976	0.358	0.528

6.2. Appendix B – Generalisation diagnostic

Table B1 - Train and test performance of the baseline decision tree at the default 0.50 threshold.

Dataset	Support	Accuracy	Precision	Recall	Churn-F1	AUC
Train	4,225	0.799	0.645	0.535	0.585	0.824
Test	1,409	0.781	0.605	0.500	0.548	0.802

6.3. Appendix C – Held-out demographic error audit

Table C1 - Held-out decision-tree error rates across gender and senior-citizen groups.

Attribute	Group	Support	Churn prevalence	Recall	False-positive rate
Gender	Female	693	0.276	0.508	0.141
Gender	Male	716	0.256	0.492	0.096
SeniorCitizen	No	1,183	0.235	0.478	0.112
SeniorCitizen	Yes	226	0.425	0.563	0.162

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (Codex 5.5)
- Anthropic Claude (Opus 4.8)

Both tools were used to assist with understanding classification and customer-churn concepts, structuring the leakage-safe PySpark workflow, evaluating missing-value and model-improvement strategies, improving academic clarity, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"How should I split this imbalanced churn dataset into training, validation and test sets, and fit the PySpark preprocessing stages without leaking information from held-out customers?"*
2. *"Contract is the decision tree's most important feature. How can I simulate 30% missingness, apply training-derived mode imputation, and compare performance on the same held-out customer IDs?"*
3. *"The decision tree achieves 78.1% accuracy but only 50.0% churn recall. Explain why accuracy is misleading here and how threshold tuning, class weighting and Random Forest affect recall, precision, F1 and AUC."*

I confirm that the use of the AI tool has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

7. References

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